Philmont Scout Ranch Geologic Synthesis for Trek 12–15

Executive Summary

Trek 12–15 is one of the better Philmont itineraries for anyone who wants to read the landscape geologically, because it does not stay in one rock package or one landform belt. The route begins in the low eastern foothill country, climbs through the central ridge-and-dike belt, reaches the Baldy intrusive and basement core, then drops back into the northern basin succession around French Henry, Pueblano, Ponil, Indian Writings, and the North Ponil dinosaur-track area before finishing near base. In practical terms, that means you are likely to see dark marine shale terrain, resistant fluvial sandstone and conglomerate, igneous dikes and intrusive bodies, mineralized fault/contact zones, mining landscapes, canyon incision, and culturally important sandstone outcrops on a single trek. The official itinerary confirms that emphasis: Baldy Mountain, Dean Skyline, Deer Lake Mesa, Baldy Mining District, French Henry, Ponil, Indian Writings, and the T-Rex Track are all explicit program or hiking highlights. fileciteturn0file0

The big geologic story told in the older Philmont guides still holds up. Philmont lies on the eastern flank of the Cimarron Range where the Great Plains meet the easternmost Southern Rockies. The eastern and lower parts of the ranch are dominated by gently east- to northeast-dipping Upper Cretaceous marine shales; the central country is strongly shaped by resistant Tertiary porphyry dikes and sheets that stand out as ridges; and the western/northwestern high country near Baldy exposes older rocks and intrusive centers associated with uplift and mineralization. That general field picture is laid out clearly in Robert Balk's trail guide and remains the most useful walker-scale framework. [1]

What modern work changes is mostly the level of detail rather than the fundamental story. Compared with the mid-century field guides, later map products and later summaries distinguish surficial deposits more carefully, refine stratigraphic nomenclature in the basin section, and tie ore bodies more explicitly to faults, dikes, and the Pierre–Raton contact zone in the Baldy-French Henry area. In other words, **PP 505 remains highly relevant for hiking interpretation**, but modern syntheses are stronger on exact contacts, ore controls, and stratigraphic correlations. That is the right way to use the older guide: trust it for landscape reading, but not as the last word on age assignments or mapping detail. [2]

For Trek 12–15 specifically, the most defensible field-ready interpretation is this: Days around Harlan, Devils Wash Basin, and the return to HQ are dominated by the Cretaceous basin margin and its shale/sandstone landscape; the Baldy days are dominated by intrusive rocks, crystalline core exposures, and structurally controlled ore; and the Ponil–Indian Writings–T-Rex segment is dominated by resistant basin sandstones and canyon incision, with paleontologic and archaeological significance layered onto the geology. Exact GPS-by-formation crossings remain **schematic** in this report because the accessible materials included the itinerary but not an accessible route KMZ or georeferenced sectional GIS track. fileciteturn0file0

Regional Framework and Revisions

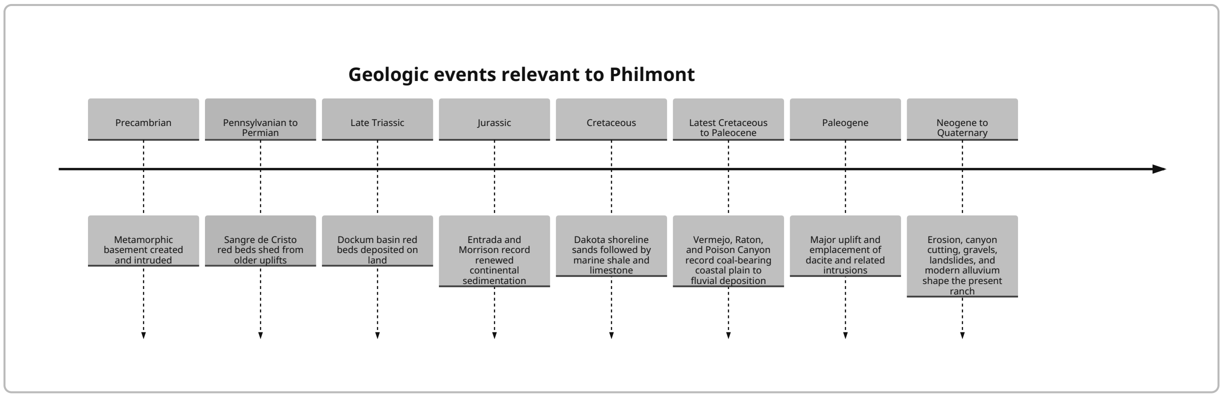
Philmont occupies the eastern slope of the Cimarron Range, which Balk described as the easternmost of the north-south Rocky Mountain ridges in this part of New Mexico. From Tooth of Time Ridge, he emphasized the juxtaposition of a relatively low area around headquarters, the Raton Basin plateau to the north, and basalt-capped plateaus to the south. He also described the eastern ranch as underlain mainly by soft Cretaceous Pierre shales, dipping gently east or northeast, while the middle part of the ranch is cut by resistant quartz-monzonite/porphyry dikes that create conspicuous ridges and cliffs. [1]

That walker-scale framework is still the right starting point for Trek 12–15. On the eastern and southeastern legs of your route, expect broad slopes, rounded hills, and muddy or “gumbo” behavior where shale dominates. In the central country, expect long, linear ridges, cliffy ledges, and prominent skyline features where porphyry sheets or dikes resist erosion. In the Baldy sector, expect a much rougher, more crystalline and mineralized landscape. North and northeast of Baldy, around French Henry and Ponil, expect more resistant fluvial sandstone and conglomeratic units of the Raton Basin succession than you see near headquarters. That is the topographic logic of the trek. [3]

Several points in the older literature deserve updating. Balk’s 1953 trail guide tentatively suggested that some of the porphyry dikes might be Miocene, which was an honest statement of uncertainty for the time. Later summaries and later map products place Philmont’s major intrusive episode more broadly in the Tertiary, and popular modern summaries of features like Tooth of Time place the dacite intrusion in the Paleogene rather than the Miocene. I would treat the older “perhaps Miocene” phrasing as outdated and the broader Paleogene/Tertiary assignment as the more usable modern interpretation. [4]

Another modernization is stratigraphic detail. Philmont’s older teaching plates emphasize a practical ranch-scale sequence: Precambrian gneiss/schist and granodiorite at depth; Sangre de Cristo red beds; Dockum Group red beds; Entrada and Morrison; Dakota; the marine Cretaceous shale and limestone interval; Trinidad, Vermejo, Raton, and Poison Canyon; then intrusive rocks, basalt, and Quaternary deposits. That remains a good field order, but later regional work across northeastern New Mexico has refined terminology within some of those packages and sharpened debates about Triassic nomenclature, especially around Dockum versus Chinle-related usage in northeastern New Mexico. For an advanced hiker, the important point is not to memorize every nomenclatural dispute; it is to understand that Philmont records repeated shifts between terrestrial basins, shallow seas, fluvial/coastal plains, and later uplift plus intrusion. [5]

PP 505 and the older trail guide also stressed that Philmont proper shows no clear evidence of classic Pleistocene glaciation such as moraines, cirques, or roche moutonnée. That remains a useful caution because hikers often overinterpret every high-country hollow as glacial. On Baldy and nearby high ground, it is more defensible to think first about structural control, differential weathering, frost action, talus, and stream incision than about glaciers. [6]



That sequence is the simplest accurate mental model to carry on trail: **basement and red beds in the west, marine Cretaceous mud and sand to the east, fluvial basin sandstones in the north, and younger intrusive rocks cutting through all of it.** [7]

Trek Stratigraphy and Field Recognition

The table below is the most useful “what am I standing on?” summary for Trek 12–15. It is intentionally field-oriented rather than encyclopedic. Because an accessible KMZ or georeferenced trek line was not specified, the “GPS extent” column is stated as a route window rather than exact coordinates. The unit assignments are based on the itinerary sequence plus Philmont geologic guide/map materials, so treat them as **schematic but operationally useful**. filecite turn0file0 [8]

Formation or rock package	Age	Typical lithology	Diagnostic field clues	Most likely Trek 12–15 occurrence	GPS extent along trek
Quaternary alluvium, terrace gravel, colluvium	Quaternary	Sand, gravel, modern wash fill, slope debris	Active creek bottoms, broad benches, rounded loose gravels, flood-prone channels	Cimarron River, Ponil drainage, South/North Ponil bottoms, near HQ approach	Exact GPS extent unspecified
Pierre Shale and related Upper Cretaceous marine shale interval	Late Cretaceous	Dark gray to black shale, bentonitic/ clayey beds, locally orange-weathering ash-rich shale	Soft slopes, bad footing when wet, sparse cliff formation, swelling clay, broad rounded topography	HQ–Harlan sector, much of the eastern return leg, parts of central country beneath dike ridges	Exact GPS extent unspecified

Formation or rock package	Age	Typical lithology	Diagnostic field clues	Most likely Trek 12–15 occurrence	GPS extent along trek
Trinidad Sandstone and Vermejo transition beds	Latest Cretaceous	Light buff sandstone with adjacent carbonaceous shale/coal-bearing beds	Resistant ledges above shale, carbonaceous fragments, transitional shoreline/ coastal-plain look	Likely north-country and Ponil sector windows; especially where sandstone cliffs begin to replace pure shale slopes	Exact GPS extent unspecified
Raton Formation	Latest Cretaceous to Paleocene	Yellow to buff sandstone, shale, local conglomerate, coal/ carbonaceous beds	Cross-bedding, mudcracks/sun cracks, resistant ribby outcrops intermixed with carbonaceous intervals	Ponil, Indian Writings, T-Rex-track sector, plus parts of the north-country return	Exact GPS extent unspecified
Poison Canyon Formation	Paleocene	Coarse yellow pebbly sandstone and conglomerate, fluvial channel deposits	Bench-forming conglomeratic outcrops, pebble lags, coarse stream-laid architecture	North-country around Pueblano/ Ponil benches and canyon rims; likely visible from overlooks even where not directly crossed	Exact GPS extent unspecified
Dacite/ porphyry dikes, sheets, plugs, laccolithic bodies	Tertiary	Light-colored porphyry, commonly with feldspar phenocrysts; massive, resistant	Straight ridges, ledges, cliff bands, discordant bodies cutting layered rocks	Dean Skyline, Head of Dean area, central ridge belt, French Henry–Black Horse–Tooth-of-Time-style ridges	Exact GPS extent unspecified

Formation or rock package	Age	Typical lithology	Diagnostic field clues	Most likely Trek 12–15 occurrence	GPS extent along trek
Crystalline core rocks and related intrusive basement complex	Precambrian basement plus older intrusive rocks exposed in uplifted core	Gneiss, schist, amphibolitic/ metamorphic rocks, granodioritic rock, pegmatitic material	Coarse crystalline texture, foliation in metamorphic rocks, lack of bedding, blocky talus	Baldy summit area and immediate high-country approaches	Exact GPS extent unspecified
Basalt mesa caprock	Neogene to Quaternary regionally	Dark dense basalt, locally columnar	Columnar joints, caprock cliffs, mesa tops	More of a visual backdrop than a dominant tread surface on this itinerary; relevant around southern/ adjacent mesas rather than the core of 12–15	Exact GPS extent unspecified

What you should actually do with that table on trail is simple. If the terrain is broad, smooth, and clay-rich, think **marine shale**. If it turns cliffy, pebbly, and channelized, think **Raton or Poison Canyon fluvial sandstone/conglomerate**. If the ridge is unusually linear and massive relative to surrounding bedded rock, think **porphyry dike or sheet**. If the rock loses bedding entirely and becomes crystalline, you are near the **Baldy core**. [9]

Simplified geologic setting of Trek 12–15 with schematic overlay and approximate profile

The figure above is deliberately schematic. It combines the official itinerary geometry with the major Philmont geologic domains recognized in the trail guide and mining literature, and it is meant to help you anticipate transitions rather than replace sectional maps. The route trend is robust even though the exact track is not GPS-precise here. filecite turn0file0 [8]

A few field-recognition rules matter more than all the rest. Pierre shale commonly makes the landscape rather than the outcrop: look for low, soft, often vegetation-mantled slopes and slick footing after rain. Raton and Poison Canyon outcrops are where you start seeing better bedforms, tougher ledges, and more obvious channel architecture. Basalt, where present, may show columnar jointing; Balk specifically drew attention to vertical columns in basalt exposures as a cooling-shrinkage pattern. Porphyry is easy to overcall, so remember this: if you can still see clear bedding, you are in sedimentary rock; if the outcrop is massive, light colored, phenocryst-rich, and cuts across layered host rocks, you are probably looking at an intrusion. [10]

Mining Geology of Baldy and French Henry

The Baldy district is where Philmont stops being just scenic geology and becomes economic geology. The 1966 New Mexico Geological Society synthesis makes the controls unusually clear. In the Ute Creek district east of Baldy, the Aztec mine — the district's major producer — is interpreted as mineralization derived from a monzonite porphyry sill below the Pierre–Raton contact, with ore movement focused along the Aztec fault and related structures. Early ore came from quartz veins in sandstone near the base of the Raton Formation and from the sandstone–Pierre contact; later, larger ore bodies were worked in shale. Reported minerals include quartz, calcite, pyrite, chalcopyrite, galena, and pyrrhotite. That is the core geologic lesson of Baldy mining: ore followed **intrusion + fault + favorable contact**. [11]

The surrounding mines show the same structural logic, but with different host geometries. The Montezuma ore occurs in pyrite finely disseminated through a quartz vein cutting sills and dark siliceous shale. The Bull of the Woods lode is an extension of the Montezuma lode. The Black Horse group carried two quartz veins a few feet wide within a monzonite dike, and Pettit reported values as high as $100/ton in places, though average values were much lower. For a trekker, the take-home point is that the district is not random: your route is crossing a network of mineralized veins, faults, and porphyry bodies that were systematically exploited because the structures concentrated fluids and ore. [12]

French Henry belongs to the same overall district story but with its own structural flavor. Pettit's summary for the Ponil district states that the French Henry mine lies on French Henry Mountain north of South Ponil Creek, and that numerous faults cut the Pierre and Raton Formations and associated dikes there. Ore occurred in the Yellow Dog, French Henry, and Black Joe fault zones and also along the Pierre–Raton contact, with especially rich values in ferruginous fault gouge. That is a textbook structural-ore setting for hikers to keep in mind: **fault gouge, contact zones, and intrusive association all reinforce one another**. [13]

The historical arc matters because it explains what you can still see. Gold discovery on Baldy triggered an 1866–1867 rush, placer operations on the west side of Baldy, rapid development of district infrastructure, and later hard-rock exploitation at Aztec, French Henry, Black Horse, and related claims. Pettit describes the "Big Ditch," placer development, and early district organization in enough detail to show that mining here quickly became an engineered landscape rather than just pick-and-pan prospecting. [14]

On Trek 12–15 the mining evidence is therefore not background scenery. Around Baldy Town, Black Horse, and French Henry, look for waste rock, altered or iron-stained material, linear excavations, adit zones, mill-site terraces, cabin remnants, tramway corridors, and changes in vegetation where tailings or disturbed ground persist. The itinerary itself confirms that Baldy Mining District interpretation is a core program element on the Baldy days and that French Henry is an intended interpretive stop. Modern summaries of French Henry also note that the mill site retains standing cabins and serves as a living-history program area, whereas the actual mine is abandoned and access is restricted. filecite turn0file0 [15]

The right field behavior around these sites is bluntly non-negotiable. Do **not** enter adits, shafts, collapsed workings, or mill ruins; do not climb waste piles or unstable cribbing; and do not pocket ore from interpretive sites. If you want to work like a field geologist, record structures and minerals from a safe distance: note whether a vein lies in sandstone, shale, or porphyry; note whether the lode is concordant with bedding or crosscuts it; and note whether limonite/hematite staining appears concentrated along faults or contacts. Those observations will teach you more than trying to carry home a specimen. [16]

Paleontology and Human Landscape

The paleontologic headline for Trek 12–15 is obvious: the itinerary ends with the T-Rex Track. The published formal significance is that the Philmont footprint was described as *Tyrannosauripus pillmorei*, a giant tridactyl tyrannosaurid-like track from the Late Cretaceous of New Mexico, discovered by Charles Pillmore and formally named by Lockley and Hunt in 1994. Popular summaries differ on some details, including the exact discovery year and reported footprint dimensions, so the safest interpretation for the field guide is this: you are looking at a famous, controversial-but-important tyrannosaurid-style footprint from a latest Cretaceous wetland or mudflat context in the North Ponil area, not at a generic “dinosaur print.” [17]

That matters for landscape reading because the track is not an isolated curiosity. It belongs to the same Raton Basin basin-margin succession that records the shift from marine Cretaceous conditions to coastal-plain, swamp, and fluvial environments at the end of the Cretaceous and into the Paleocene. The same broad depositional setting that produced coal-bearing and carbonaceous beds also produced periodically muddy, exposed surfaces capable of preserving tracks. So when you reach the North Ponil/T-Rex-track area, look not just for the track itself but for the surrounding sedimentary logic: bedding surfaces, changes in grain size, evidence of channel migration, and the alternation between resistant and recessive units. [18]

Indian Writings is the main archaeological stop on the itinerary, which explicitly frames it as an Ancestral Puebloan archaeology program. Geologically, that pairing makes sense. Petroglyph localities concentrate where durable, exposed rock surfaces are available and where the outcrop stands high enough above the valley floor to remain visible and meaningful over time. On Trek 12–15, Indian Writings is best treated as a sandstone-cliff interpretive stop where geology created the surface and human history gave it meaning. The field ethic here is even stricter than at the mining sites: no touching, wetting, chalking, tracing, or close-up abrasion of rock art surfaces. filecite turn0file0 [19]

The human-history framework for the trek runs through the Maxwell Land Grant. The grant began as the 1841 Beaubien-Miranda land grant, later became associated with Lucien Maxwell, and then shaped settlement, mining, ranching, and conflict across the whole district. Baldy Town, Elizabethtown, Cimarron, Ponil-country settlement, and much of the later Philmont landscape all sit inside that legal and economic history. In practical terms, the grant history explains why this trek’s geology is inseparable from roads, ditches, mills, cabins, and ghost-town remnants: the rocks were monetized early and aggressively. [20]

A good sample panoramic interpretation point is **Baldy summit**. Looking west, think placer systems and the Moreno Valley, where erosion redistributed gold from higher hard-rock sources into workable gravels. Looking east and southeast, think Ute Creek, Black Horse, and the contact/fault-controlled hard-rock mines. Looking north and northeast, think French Henry Mountain and the north-basin sedimentary section. Looking south, think the transition back into the central dike belt and the broader Philmont ridge-country skyline. This is partly a direct reading of the mining districts and partly an inference from the regional geologic framework, but it is the right way to teach yourself to see Baldy as a map, not just a summit. [21]

Trek 12–15 Day-by-Day Analysis

The itinerary below is the most useful version for a hiker: what you are likely to cross, what to look for, and one or two field exercises worth actually doing. Because the route geometry available here is itinerary-scale rather than GIS-precise, these are best read as **segment interpretations**, not surveyed contact calls. filecite turn0file0

Days around Harlan and Devils Wash Basin

From Camping HQ to Harlan, the route likely begins in the eastern shale country, where Upper Cretaceous marine mudrock dominates the lower relief. Expect soft hills, drainage incision rather than cliff-forming outcrop, and possible local influence from the central dike belt as you move westward. The right question on this segment is: **why do some ridges stand up while others just dissolve into rounded slopes?** Usually the answer is either a resistant sandstone cap or an intrusive body. filecite turn0file0 [22]

From Harlan to Devils Wash Basin, watch for the landscape becoming more structurally organized. If you begin seeing straighter ridges, steeper slopes, or isolated gray ledges, you are likely approaching the porphyry-controlled central country described by Balk. This is a good day for a simple **strike-and-slope exercise**: sketch the skyline and ask whether the ridges follow bedding, cross bedding, or ignore bedding entirely. If they ignore bedding and hold a line anyway, intrusion is the better hypothesis. filecite turn0file0 [23]

Days around Cimarron River and Santa Claus

The Devils Wash Basin to Cimarron River leg is a classic descent from higher structural ground into an actively incised drainage. The geologic lesson is differential erosion: weaker rocks and fracture zones open valleys, while stronger units or intrusive ribs hold the shoulders. Around the Cimarron River, focus on modern alluvium, terrace surfaces, and the way the drainage exploits weaker rock compared with nearby ridges. A useful field exercise is to stand where you can compare the floodplain, the first bench above it, and the enclosing slopes, then decide what is active channel, what is abandoned terrace, and what is bedrock valley wall. filecite turn0file0 [24]

The Cimarron River to Santa Claus climb is the transition back toward the western and more structurally complicated part of the ranch. If the tread passes from broad shale-derived slopes into redder or more ledgy rock, you may be brushing older sedimentary windows on the way toward the core, though the exact crossings need the GIS line to pin down precisely. The important observation is the **change in rock competence**. A second useful exercise here is to compare float fragments at the bottom and top of the day’s climb: if the float becomes coarser, more crystalline, or more iron-stained upslope, you are moving toward the mountain core and/or mineralized zone. filecite turn0file0 [24]

Baldy Town layover and Black Horse

Santa Claus to Baldy Town is where the geology gets serious. This segment climbs into the Baldy intrusive and crystalline core and toward the Ute Creek district mines. Expect a jump in relief, more blocky rock, more obvious structural control, and increasing evidence of historic mining. The itinerary’s “Head of Dean” and “Baldy hike prep” are not just programmatic; they put you in the zone where porphyry, faulting, and basement exposure dominate the landscape logic. filecite turn0file0 [25]

The Baldy layover day is your best chance to do real economic-geology observation without adding mileage. On the Black Horse/Baldy side, look for quartz vein material, iron staining, altered wall rock, and the contrast between host sedimentary rocks and intrusive rock. Do **not** think only in terms of “gold here.” Think instead in terms of the plumbing system: what fractures existed, what host rocks were favorable, and why fluids deposited ore there and not fifty feet away. A good field exercise is a **contact-control notebook sketch**: draw host rock on one side, intrusive on the other, and note where staining, quartz, and mine workings cluster. filecite turn0file0 [26]

French Henry, Pueblano, and Elkhorn

Baldy Town to Pueblano via French Henry is one of the most educational segments of the entire trek. You leave the immediate Baldy core and re-enter the structurally mineralized sedimentary basin edge, where faults, dikes, and the Pierre–Raton contact control ore at French Henry. This is the day to compare Aztec-style contact/fault mineralization and French Henry-style fault-zone/contact mineralization as expressions of the same regional intrusive system. At French Henry, focus on structures you can still see safely: fault traces, mill-site topography, altered rock, and the relationship of worked ground to the ridge. filecite turn0file0 [27]

Pueblano to Elkhorn is a transition day into the northern basin section. The landscape should start to look more like resistant fluvial sandstone and conglomerate country than like core-country intrusion country. This is a good place to do a **channel-sandstone exercise**: look for pebbly zones, stacked ledges, cross-bedding, or erosive basal surfaces in more resistant units, and compare them with the smoother, recessive slopes of finer-grained beds. If you see a sandstone body that clearly forms a bench or rim, ask whether it looks beach-like, channel-like, or simply massive and intrusive. filecite turn0file0 [28]

Ponil, Indian Writings, and the finish

Elkhorn to Ponil and then Ponil to Indian Writings bring you into the canyon and benchland country where Raton and Poison Canyon-type units are the most useful working hypothesis. This is the place to look carefully for sedimentary structures rather than mining structures: cross-bedding, pebble lags, mudcracks, erosional channel contacts, and changes from carbonaceous intervals into clean sandstone. The old Philmont material specifically highlighted cross-bedded yellow sandstone and sun-cracked sandstone at Ponil Creek as classic observations. Even without that exact outcrop in hand, the method is still right: **read the sandstone bodies as former channels and floodplains**. filecite turn0file0 [18]

Indian Writings is both a cultural and geologic stop. Treat the sandstone face as a long-lived, durable recording surface that survived because the rock was physically competent enough to remain exposed. A useful field exercise here is **landscape placement**: ask why the site is here, not a hundred yards away. Usually the answer involves visibility, access, shelter, and the geometry of resistant outcrop. filecite turn0file0

From Indian Writings back to Camping HQ via the T-Rex Track and Six Mile Gate, the interpretive emphasis shifts to final basin history. The track is the paleontologic prize, but the broader lesson is the same one you have been building all trek: the route back to base is not a random corridor; it is a traverse through the late basin-fill and margin sequence that records the environments dinosaurs and later rivers both occupied. Your final field exercise should be the simplest one of the whole trip: write one paragraph answering, in plain English, **how Philmont went from sea to swamp to river plain to uplifted mountain front**. If you can do that from memory at Six Mile Gate, the guide has done its job. filecite turn0file0 [29]

Field Methods and Quick-Reference Page

A field guide for this trek should bias toward things you will really use in the rain, at altitude, and with tired legs. The useful kit is a small waterproof notebook, pencil, hand lens, baseplate compass, your sectional map, a thin ruler, a phone for photos only if power budget allows, and a quart freezer bag or map case. I would not carry a rock hammer, acid bottle, or specimen bags on a Philmont trek; they add weight, create temptation, and are poorly matched to the ethics of archaeological, paleontological, and mine-interpretive stops. That recommendation is an inference from the nature of the sites on this itinerary and from mine/archaeology safety concerns. filecite turn0file0 [30]

Hand-sample identification at Philmont should stay minimalist. Quartz is hard, glassy, and lacks cleavage. Calcite is softer and cleaves, but on trek you are unlikely to do formal acid tests. Pyrite is brassy and cubic or granular; chalcopyrite is brassier and typically more tarnished; galena is dense, metallic, and lead-gray; limonite/hematite show up as rusty to red-brown weathering in many mine settings. With no destructive tools, the best method is visual comparison: luster, cleavage, crystal habit, relative hardness against a fingernail or knife spine only on loose float, and context in the outcrop. If it comes from a vein in a mine district, record its setting first and the mineral name second. [31]

Outcrop reading should follow the same order every time. First ask whether the rock is bedded or massive. Second ask whether the landform is slope-forming or cliff-forming. Third ask whether structures are depositional, intrusive, or tectonic. Fourth ask what the rock is doing in the landscape — channel, ridge former, caprock, fault zone, contact zone, talus source, or valley fill. If you can answer those four questions, you are already doing competent field geology. [32]

For a laminated quick-reference page, I would include exactly this content:

Front side	Back side
Formation quick ID: Pierre shale, Trinidad/Vermejo, Raton, Poison Canyon, porphyry, crystalline core	Ore/mineral quick ID: quartz, pyrite, chalcopyrite, galena, limonite/hematite, calcite
Sedimentary structures: cross-bedding, mudcracks, channel base, pebble lag, terrace bench	Structural cues: dike vs sill, fault gouge, contact zone, joint set, ridge controlled by intrusion
Landscape cues: shale slope, sandstone bench, porphyry ridge, canyon floor alluvium	Safety/ethics: no mine entry, no artifact touching, no fossil collecting, no outcrop hammering
Daily log prompts: rock, structure, landform, human use	Weather prompts: lightning risk on Baldy, flash-flood awareness in Ponil/Indian Writings country

The short version of the entire report is this: **Trek 12–15 is a traverse from shale basin margin, through intrusive ridge country, into a mineralized mountain core, then back through sandstone canyon country loaded with mining history, archaeology, and the most famous dinosaur track at Philmont.**

PP 505 is still the right backbone for the story; later work mostly adds sharper mapping, better ore-control language, and more modern stratigraphic nuance. filecite turn0file0 [33]

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[1] [2] [3] [4] [5] [6] [7] [8] [9] [10] [22] [23] [24] [29] [32] [33] https://geoinfo.nmt.edu/publications/openfile/downloads/0-99/22/ofr_22.pdf
https://geoinfo.nmt.edu/publications/openfile/downloads/0-99/22/ofr_22.pdf

[11] [12] [13] [14] [21] [25] [26] [27] [31] https://nmgs.nmt.edu/publications/guidebooks/downloads/17/17_p0069_p0075.pdf
https://nmgs.nmt.edu/publications/guidebooks/downloads/17/17_p0069_p0075.pdf

[15] [16] [30] https://en.wikipedia.org/wiki/French_Henry_mine
https://en.wikipedia.org/wiki/French_Henry_mine

[17] https://en.wikipedia.org/wiki/Tyrannosauripus
https://en.wikipedia.org/wiki/Tyrannosauripus

[18] https://en.wikipedia.org/wiki/Raton_Formation
https://en.wikipedia.org/wiki/Raton_Formation

[19] https://en.wikipedia.org/wiki/Philmont_Scout_Ranch
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[20] https://en.wikipedia.org/wiki/Maxwell_Land_Grant
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[28] https://en.wikipedia.org/wiki/Vermejo_Formation
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